NeuroGuard AI - architecture as implemented (app.py, core/*.py, mcp_server/neuroguard_mcp_server.py)

All LLM roles: gpt-4o-mini, temperature 0, OpenAI-compatible Vocareum endpoint | contract v1.2.0 | colour: blue UI, red guard, yellow LLM agent, green deterministic code, purple MCP, cyan observability, grey data

Streamlit UI (app.py)
profile form, note, audience, consent, RAG mode | lab report (image / PDF) live progress | 13 result tabs incl. Context-window inspector Knowledge-graph coverage tab Cohort batch tab | approve/reject .md/.pdf, receipt, OTLP export

Input guardrails (core/guardrails.py)
PHI redaction -> prompt injection -> BE FAST emergency (9-1-1) -> dosing -> out of scope + profile validation | no LLM multi-modal intake (partial): core/multimodal.py vision / pypdf

Run contract (harness_contract.json)
SHA-256 hashed before first token 36 LLM | 32 tools | 110k tok | 300 s per-agent permissions, 12 checks context, confidence, pricing, learning, batch policies

Planner (LLM)
MCP episode_recall: similar past episodes + learned hints -> structured Plan (ordered steps, watch-outs, rationale) planner -> executor pattern

Orchestrator (LLM, core/agents.py)
executes the plan: next agent, skill, instruction, reason (structured) code overrides: eligibility, FINISH only after review | 9 turns, 2 visits/agent, 1 revision | every route -> decision reasoning log

CaseState (LangGraph)
short-term memory per run: profile, plan, learning, intake, analyst, risk, evidence, coverage, retrieved, draft, self_critique, review, route, trace[]

blocked -> refusal / emergency message (+ signed receipt, 0 LLM calls)

episode_recall

Task + skill

A2A-style message bus (core/a2a.py): Agent Cards, Task -> Message -> Artifact, inproc:// endpoints (no network)

orchestrator FINISH

Intake
skill: note-screening
MCP: screen_note, memory_read (BE FAST signs, prior cases)
ReAct loop

Risk Analyst
skill: cohort-comparison
MCP: predict_risk, explain_factors, cohort_stats, calculator
+ model://risk-card | ReAct

Evidence Researcher (3 RAG modes)
classical: fixed semantic_search per topic BEFORE the LLM, docs injected
agentic (default): LLM calls search_guidelines + graph_traverse
agentic search: files, skills, SQL, web, memory, shell | coverage -> re-retrieve (max 2) | context_window

Clinical Writer (reflection)
skills: risk-communication | patient-friendly-summary
prompt clinical_report; evidence packed to 3500 tokens; learned hints
self-critique: checks + quote-verified LLM critique, 1 revision

Safety Reviewer
skill: red-team-review
12-check invariant verifier (code) + LLM red team (prompt red_team_review), quote-confirmed | revise x1

Tool-using agents run a ReAct loop: Thought -> LLM tool_calls (bind_tools) -> pre-tool-use hook (allow/deny, clamp, pin) -> MCP Observation (PHI-redacted, truncated to 1500 tokens) -> context (compacted at 12000), max 4 rounds

load_body()

Permission engine + pre-tool-use hook
least privilege per agent (bind_tools + re-check); arg clamps (k, limit, hops, expr length); tool budget; session_id and learned boost_factors pinned by harness; memory_write needs consent; shell argv allowlist, URL SSRF guard, read-only SQL, own-catalog skills only (re-checked in the server) | audit log

stdio JSON-RPC

MCP server: official mcp SDK (FastMCP), stdio subprocess
19 tools: predict_risk, explain_factors, cohort_stats, screen_note, search_guidelines, graph_traverse, calculator, memory_read, memory_write, episode_recall, episode_write, semantic_search, search_files, load_skill, execute_query, web_search, web_fetch, get_memory, shell
5 resources: guideline://index, model://risk-card, dataset://stroke-card, policy://harness-contract, graph://knowledge-graph
+ 1 resource template: guideline://{chunk_id} | 2 prompts: clinical_report, red_team_review
client: core/mcp_client.py + langchain-mcp-adapters (sys.executable)
OPENAI / NG_RECEIPT / GH keys stripped from the server's environment

write

Finalize gate + confidence policy gate
re-runs the 12 checks on the final draft
confidence = weighted calibration, cohort n, topic coverage, verifier, red team, grounding
< 0.4 refuse | < 0.7 clinician review
clinician sign-off if high risk / red flags
memory_write (consent) + episode_write (harness)

Skills (skills/, core/skills.py)
6 SKILL.md; catalog = name + description only; full body injected when the orchestrator selects it (progressive disclosure)

Context windows (core/context.py)
tiktoken o200k_base counts
truncate / compact / pack evidence
window 128k, budgets per agent
slot inspector: system, skills, tools, memory, history, user, calls, docs

Decision log + observability (core/harness.py, observability.py)
RunContext.decide(): actor, options, chosen, rationale, evidence | one span per LLM / MCP call -> OTLP-JSON export

Cost / token economics
usage per call x list price
USD 0.15/1M in, USD 0.6/1M out
per-agent cost; budgets cap spend

Signed receipt (core/pipeline.py)
HMAC-SHA256 over contract, server, inputs, report, trace, decision-log hashes + audit head, cost, confidence | clinician decision re-signs it and is written back to episodic memory

P1 data + P2 statistics
Stroke Prediction Dataset (Kaggle, 5,109 rows)
core/data.py cleaning log
core/stats.py: chi-square, Welch t, odds ratios, Wilson CIs

P3 risk model (core/model.py)
LR / RF / GBM, 5-fold CV
served: calibrated LR
exact log-odds explanations
threshold: OOF recall >= 0.80
calibration curve -> confidence

P4 notes (core/notes.py)
CDC BE FAST rules
TF-IDF + LR on MTSamples (CC0)
advisory neuro-note probability

Hybrid RAG + vectors (core/rag.py, vectors.py)
30 CDC / NHLBI excerpts
sparse TF-IDF 1-2 gram (0.6)
+ dense 24-d LSA vectors (0.4)
+ topic boost + learned boost (cap 0.06); modes sparse/dense/hybrid

Graph RAG (core/graph_rag.py)
networkx: 49 nodes (10 factor, 9 condition, 30 chunk), 144 edges
weighted multi-hop traversal
factor -> chunk, factor -> condition <- chunk, with paths

Long-term + episodic memory
SQLite: core/memory.py case summaries (consent) | core/episodes.py one de-identified episode per run -> learning loop: whitelisted hints + bounded boosts

Scale (partial): core/batch.py runs a cohort with asyncio.Semaphore(3) + one shared token bucket (120 LLM calls/min) awaited before every LLM call. Honest scope: A2A is in-process; public educational data; not a medical device.

Repo: https://github.com/sandipdey2/neuroguard-agentic-rag | Live demo: https://neuroguard-agentic-rag-4ywylkye6e2rgzavdb5xrb.streamlit.app/